5V Input 3A 1.5MHz Synchronous Step-Down DC/DC Converter

DESCRIPTION

The MT8133 is a 1.5MHz, 3A constant on-time (COT) controlled synchronous step-down converter. It can operate with input voltage from 2.5V to 5.5V and provide output range from 0.6V to input level, thanks to its 100% duty cycle operation. The constant on-time control scheme simplifies loop compensation and offers excellent load transient response. MT8133 consumes extremely low 18μA quiescent current hence achieves superior light load efficiency. The high gain error amplifier in the control loop provides excellent load and line regulation. Proprietary adaptive on-time helps MT8133 to achieve nearly constant switching frequency across load range. MT8133 has cycle-by-cycle current limit and hiccup mode to protect over-load or short circuit fault conditions.

MT8133 is available in low profile 8 leads DFN 2mm x 2mm and SOT23_5L/6L packages.

FEATURES

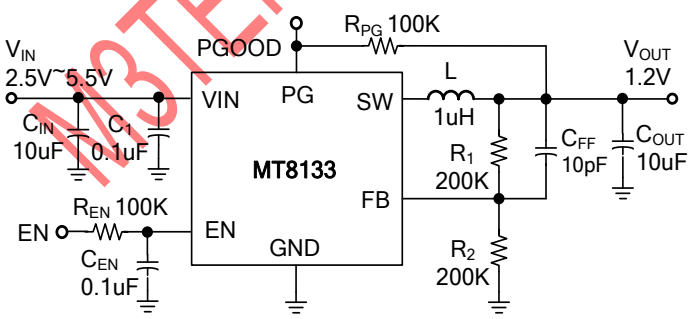
- Wide Input Range from 2.5V to 5.5V
- 3A Continuous Output Current
- Proprietary Fast Transient Constant On Time Architecture Stable with low ESR Ceramic Output Capacitors
- +/- 2% 0.6V Feedback Voltage
- 1.5MHz Switching Frequency
- 18μA Low Quiescent Current
- Up to 95% Efficiency
- 100% Duty Cycle Operation
- Built-in 85mΩ/52mΩ(DFN2X2) Power Switches
- Internal 1msec Soft-Start
- Cycle-by-cycle Current Limit Protection
- Over-Load and Short Circuit Hiccup Mode
- Open Drain Power Good Indication
- Output Discharge
- Thermal Shutdown Protection
- Available in Small DFN2x2_8L & SOT23_5L/6L
- Pb-Free RoHS Compliant

APPLICATIONS

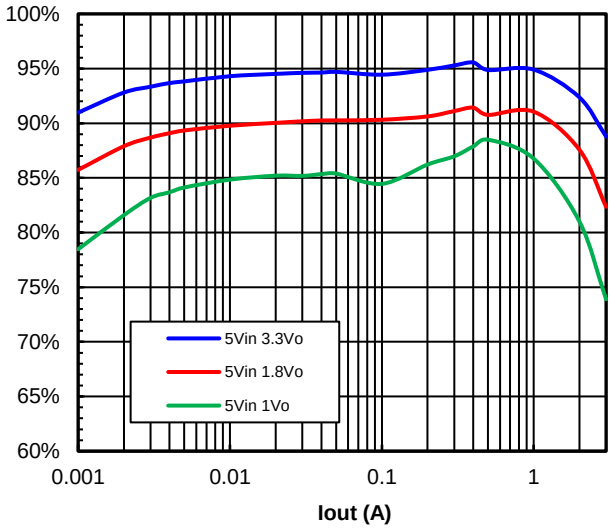
- Solid-State and Hard Disk Drives
- Smart Phone and Tablets
- WiFi RF Moudules
- DC/DC Micro Modules

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Typical Applications



Efficiency

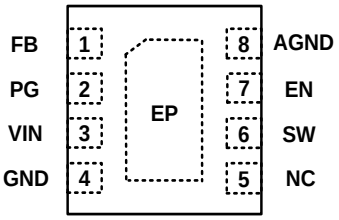


Ordering Information

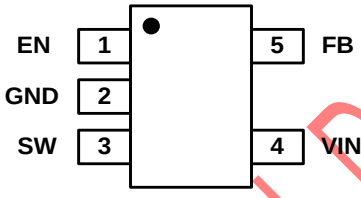
Part No.	Marking	Temp. Range	Remark	Package	MOQ
MT8133NDAR	8133 YWxx	-40°C ~+85°C	Adjustable Vout	DFN2x2_8L	3000/Tape & Reel
MT8133NSBR	8133 YWWxx	-40°C ~+85°C	Adjustable Vout	SOT23_5L	3000/Tape & Reel
MT8133NSCR	8133 YWWxx	-40°C ~+85°C	Adjustable Vout	SOT23_6L	3000/Tape & Reel

Note: Y: Year, W: Week, xx: Manufacture Code

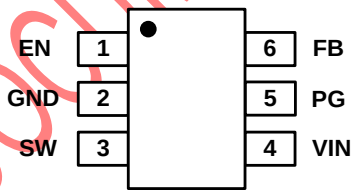
Pin Configuration



DFN2x2_8L



SOT23_5L



SOT23_6L

Pin Description

DFN2x2_8L Pin No.	SOT23_5L Pin No.	SOT23_6L Pin No.	Symbol	Description
1	5	6	FB	Voltage Feedback Input. Connect a resistor divider between output and FB to program the output voltage. VFB is regulated to 0.6V.
2	-	5	PG	Power Good Open-drain Output. Connect a 100kΩ pull-up resistor to V_{IN} or V_{OUT}.
3	4	4	VIN	Input Supply Voltage
4, EP	2	2	GND	Ground
5	-	-	NC	No connection
6	3	3	SW	Power Switch Node
7	1	1	EN	Don't float this pin. This pin has a pull-down resistor of typically 1MΩ to GND. • Drive EN above 1.05V to turn on the converter • Drive EN below 0.4V to turn off the converter and discharge output
8	-	-	AGND	Analog Ground

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Absolute Maximum Rating (Reference to GND) (Note1)

V_{IN} ..... -0.3V to +6.5V
V_{SW} ..... -0.3V to V_{IN} +0.3V
Dynamic V_{SW} in 10ns Duration ..... -2V to V_{IN} +2V
The other Pins ..... -0.3V to +6.5V
Junction Temperature Range ..... -40°C to +150°C
Storage Temperature Range ..... -65°C to +150°C
Lead Temperature (Soldering 10s) ..... 260°C
ESD ..... Class 2

Recommend Operating Conditions (Note2)

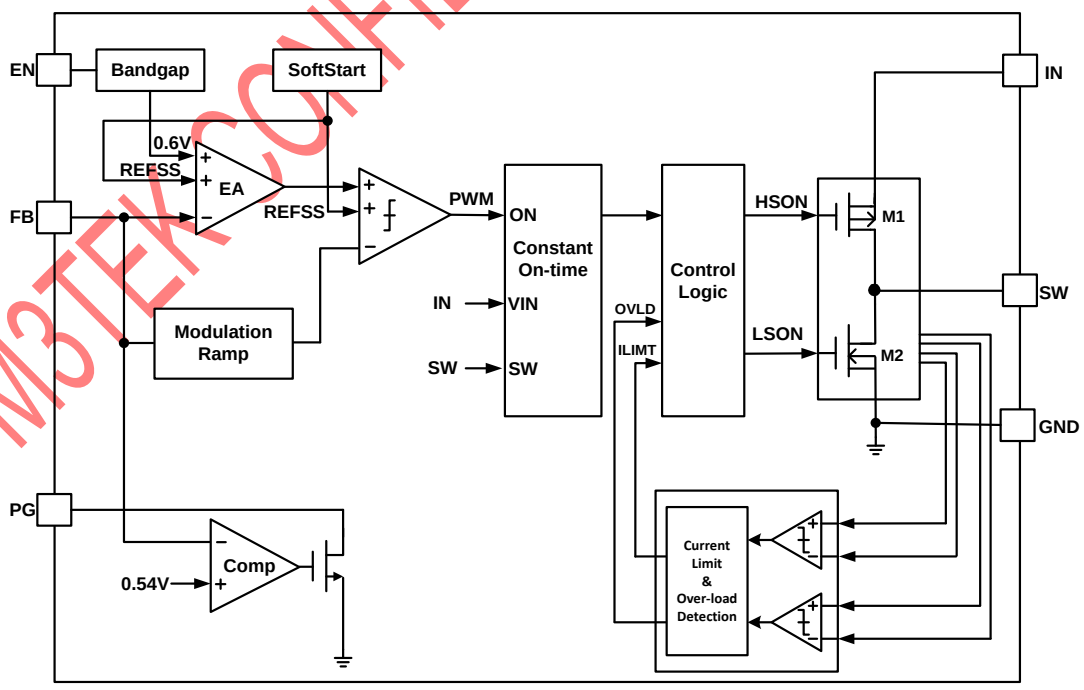
Input Voltage (V_{BAT}) ..... +2.5V to 5.5V
Output Voltage (V_{OUT}) ..... +0.6V to V_{IN}
Operating Temperature Range ..... -40°C to +85°C

Thermal information (Note3, 4)

Maximum Power Dissipation(T_A=25°C)
DFN2x2_8L 1.67W
SOT23_5L/6L 1.00W
Thermal Resistance(θ_{JA}) DFN2x2_8L ..... 75°C/W
Thermal Resistance(θ_{JC}) DFN2x2_8L ..... 40°C/W
Thermal Resistance(θ_{JA}) SOT23_5L/6L.....125°C/W
Thermal Resistance(θ_{JC}) SOT23_5L/6L . .....66°C/W

Note (1): Stress exceeding those listed “Absolute Maximum Ratings” may damage the device.
Note (2): The device is not guaranteed to function outside of the recommended operating conditions.
Note (3): Measured on JESD51-7, 4-Layer PCB.
Note (4): The maximum allowable power dissipation is a function of the maximum junction temperature T_{J_MAX}, the junction to ambient thermal resistance θ_{JA}, and the ambient temperature T_A. The maximum allowable continuous power dissipation at any ambient temperature is calculated by $P_{D_MAX}= (T_{J_MAX}-T_A)/\theta_{JA}$. Exceeding the maximum allowable power dissipation will cause excessive die temperature, and the regulator will go into thermal shutdown. Internal thermal shutdown circuitry protects the device from permanent damage.

Functional Block Diagram



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Electrical Characteristics

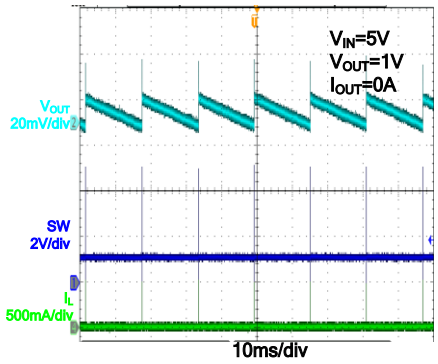
T_A = 25°C, V_{IN}=5V, unless otherwise noted. Typical values are at V_{IN} = V_{EN} = 5V.

PARAMETER	TEST CONDITIONS		MIN	TYP	MAX	UNIT
Input Voltage Range V_{IN}			2.5		5.5	V
Shutdown Current	V_{EN} = 0V, V_{IN} = 5.5V			0.1	1	μA
Input Under Voltage Lockout Threshold	V_{IN} Increasing		2.2	2.35	2.5	V
Input Under Voltage Lockout Hysteresis				250		mV
Quiescent Current I_Q	V_{FB} = 0.63V			18	24	μA
Feedback Voltage V_{FB}			588	600	612	mV
Feedback Current I_{FB}			-50	1	+50	nA
HS Switch Current Limit				4.5		A
HS Main Switch On Resistance	V_{IN}=5V	SOT23_5L(6L)		83		mΩ
		DFN2X2		85		mΩ
LS Synchronous Switch On Resistance	V_{IN}=5V	SOT23_5L(6L)		63		mΩ
		DFN2X2		52		mΩ
HS Leakage Current	V_{IN} = 5.5V, V_{EN} = V_{SW} = 0V			0.1	2	μA
LS Leakage Current	V_{IN} = V_{SW} = 5.5V, V_{EN} = 0V			0.1	2	μA
Min Off Time				65		ns
PWM Switching Frequency	I_{IOUT} = 1A			1.5		MHz
PGOOD Output Low Voltage	V_{FB} =0.5V, sink 1mA			0.2	0.3	V
PGOOD Output Leakage Current	V_{FB} = 0.63V , V_{PGOOD} = V_{IN} = 5.5V			0.01	0.2	μA
PGOOD Under Voltage Rise Threshold	V_{FB} ramp up from under voltage			-10		%
PGOOD Under Voltage Fall Threshold	V_{FB} ramp down from regulation			-15		%
PGOOD Delay	PGOOD going Low to High			30		μs
EN On Threshold	V_{EN} ramp up		1.05			V
EN Off Threshold	V_{EN} ramp down				0.4	V
VOUT Discharge	VIN=5V			50		Ω
EN Internal Pull Down Resistor				1000		kΩ
Thermal Shutdown				160		℃
Thermal Shutdown Hysteresis				30		℃

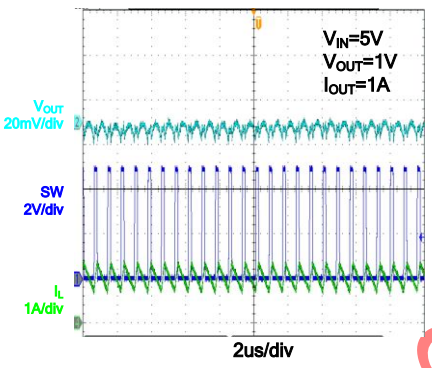
TYPICAL PERFORMANCE CHARACTERISTICS

C_{IN}=10uF, C_{OUT}=10uF, C_{FF}=10pF, L=1uH, T_A=+25℃

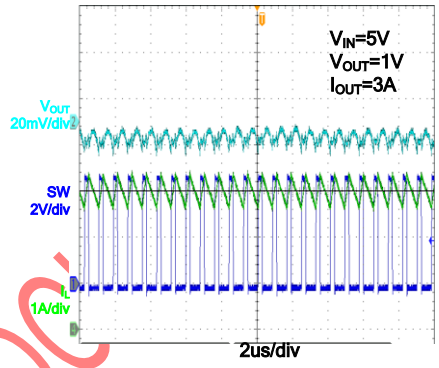
Steady State Test



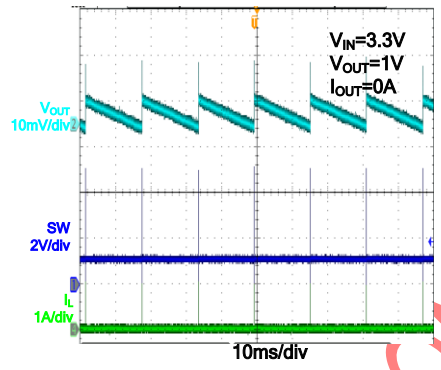
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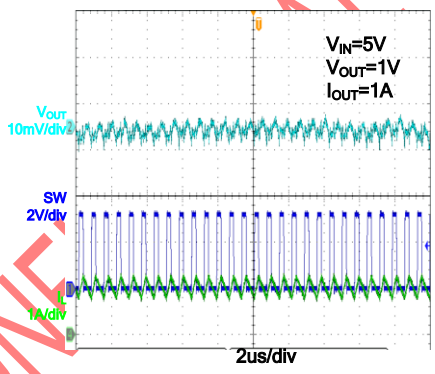
Steady State Test



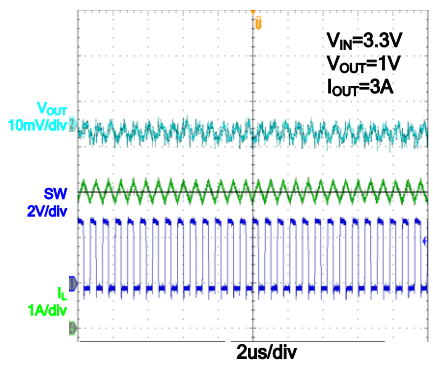
Steady State Test



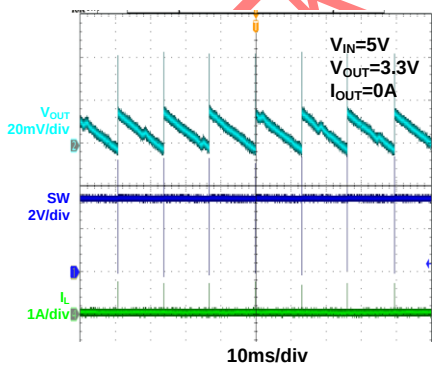
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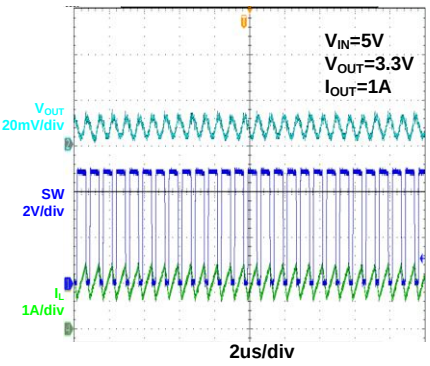
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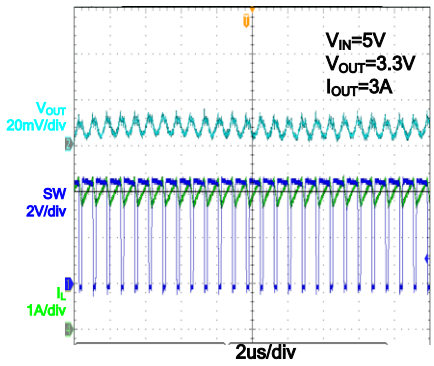
Steady State Test



Steady State Test



Steady State Test

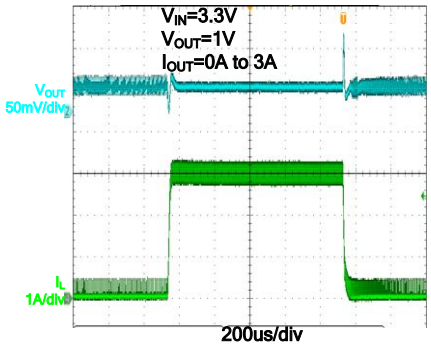


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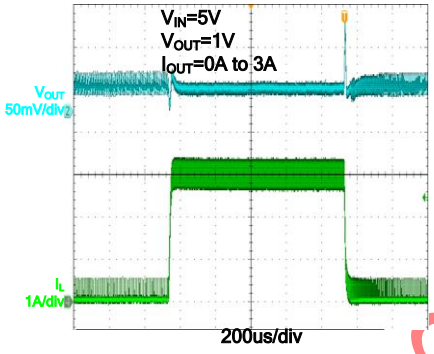
TYPICAL PERFORMANCE CHARACTERISTICS

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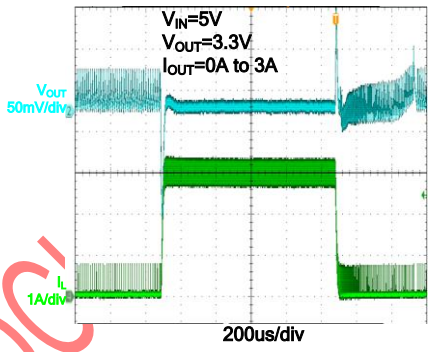
Load Transient Response



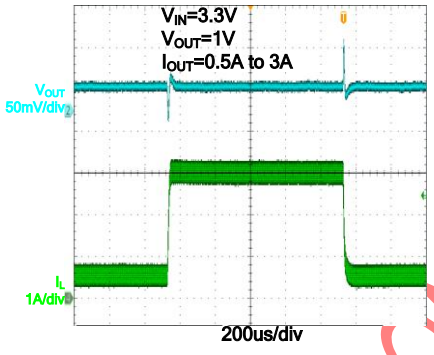
Load Transient Response



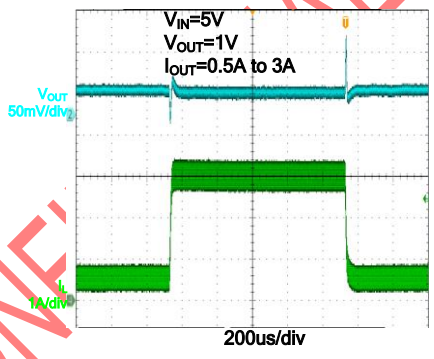
Load Transient Response



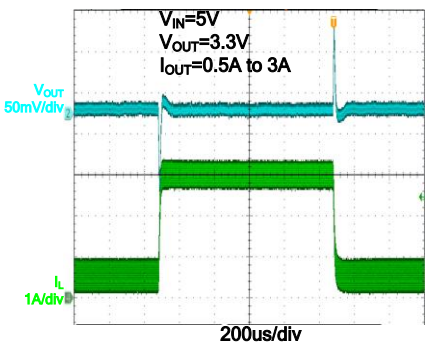
Load Transient Response



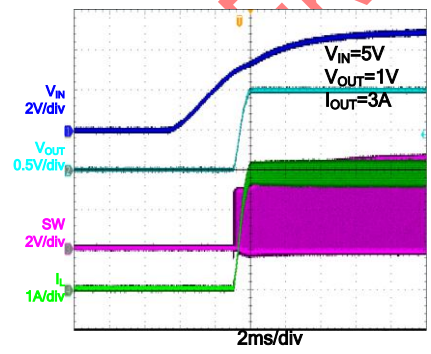
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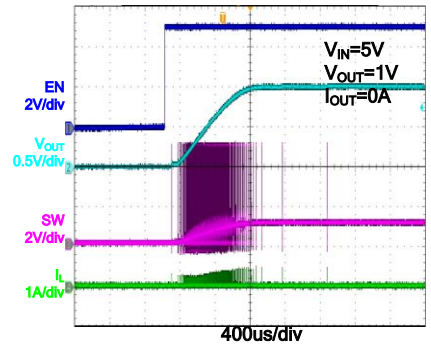
Load Transient Response



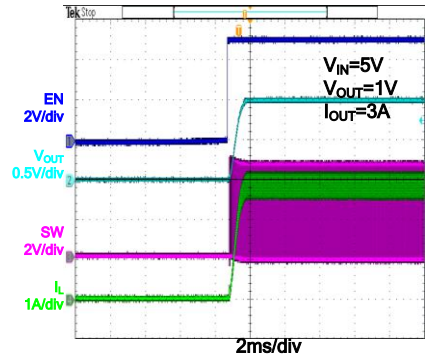
Vin Power On



EN On



EN On

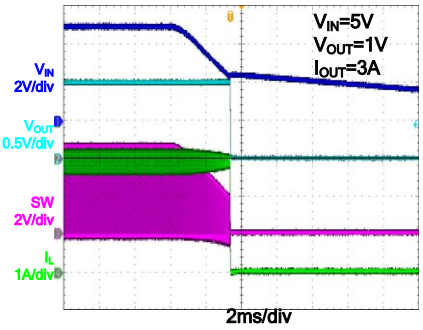


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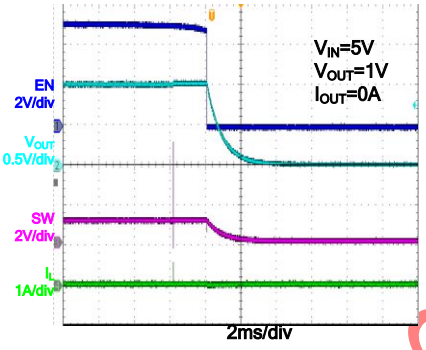
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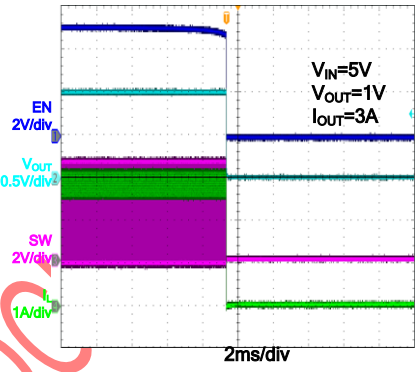
Vin Power Off



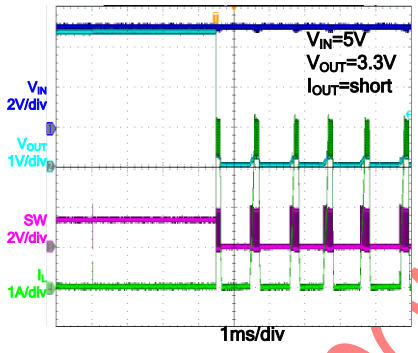
EN Off



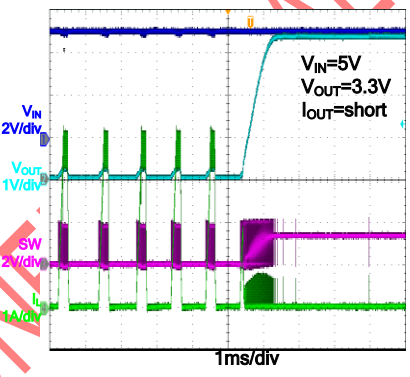
EN Off



Short Circuit Entry

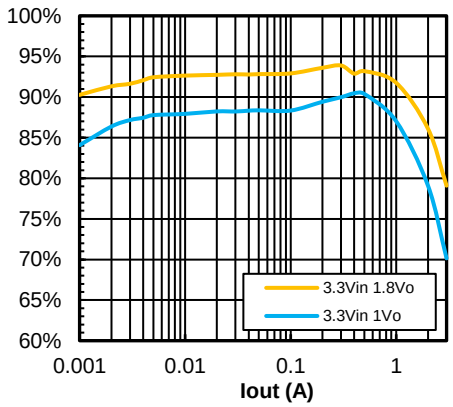


Short Circuit Recovery



Efficiency of MT8133NDAR

DFN2x2_8L



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Theory of Operation

MT8133 is a constant on-time control synchronous step-down converter that offers excellent transient response over a wide range of input voltage. It achieves superior light-load efficiency with extremely low quiescent current.

Constant On-time Control

Constant on-time control step-down converters turn on HS immediately when FB droops below reference. The HS is turned on for a pre-determined period (on-time) of time to ramp up the inductor current, and then the LS will be turned on to ramp down the inductor current. The cycle repeats itself if FB droops below reference again. MT8133 uses proprietary technique to take into account the load current impact and adjusts the on-time accordingly to achieve a constant switching frequency over entire load current range.

For MT8133, the on-time is approximately:

$$T_{ON} = \frac{V_{OUT}}{V_{IN}} \cdot 0.66\mu$$

Due to its immediate response on FB voltage droop and simplified loop compensation, constant on-time offers a superior transient response compare to traditional fixed frequency PWM control step-down converters.

Light Load Operation

In light load condition where the converter operates in discontinuous mode, MT8133 cuts down its quiescent current to as low as 15uA and achieves excellent light load efficiency.

Enable

When input voltage is above the Under-Voltage Lock-out threshold, MT8133 can be enabled by pulling the EN pin to above 1.05V. MT8133 is disabled if the EN pin is pulled below 0.4V. This pin has a pull-down resistor of typically 1MΩ to GND.

Soft Start

MT8133 has built-in soft start of 1ms. During the soft start period, output voltage is ramped up linearly to the regulation voltage, independent of the load current level and output capacitor value.

Current Limit and Hiccup Mode

MT8133 has cycle-by-cycle HS current limit protection to prevent inductor current from running away. Once HS current limit is triggered, MT8133 will turn on LS and wait for the inductor to drop down to a pre-determined level before the HS can be turned on again. If this current limit condition is repeated for a sustained long period of time, MT8133 will consider it over-load or short circuit. Either way, MT8133 will enter hiccup mode, where it stops switching for a pre-determined period of time before automatically re-try to start up again. It always starts up with soft-start to limit inrush current and avoid output overshoot.

Power Good Indication

MT8133 has open drain Power GOOD indicator PGOOD pin. When PGOOD is connected with external pull up resistor, it will be pulled up if output voltage is higher than 90% of regulation, otherwise PGOOD is pulled down by the internal open drain NMOS.

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Application Information

Setting the Output Voltage

External feedback resistors are used to set the output voltage. Refer to typical application circuit on page1, the top feedback resistor R1 has some impact on the loop stability, so its recommended range is between 100kΩ-300kΩ. For any chosen R1, the bottom feedback resistor R2 can be calculated as:

$$R_2 = \frac{R_1}{\frac{V_{OUT}}{0.6} - 1}$$

Inductor Selection

The recommended inductor value for MT8133 is between 0.33uH to 1uH. Usually the inductor value is chosen to satisfy a desired ripple current:

$$L = \frac{V_{OUT} \cdot (V_{IN} - V_{OUT})}{V_{IN} \cdot f_{SW} \cdot \Delta I}$$

where ΔI is the inductor ripple current.

With the chosen ΔI, the peak inductor current will be:

$$I_{PK} = I_{LOAD} + \frac{1}{2} \cdot \Delta I$$

Input Bypass Capacitor Selection

The input current to the step-down converter is discontinuous with very sharp edges, therefore an input bypass capacitor is required. For best performance, it's recommended to use low ESR ceramic capacitors and place them as close to the input pin as possible. For lowest temperature variations, use X5R or X7R dielectric ceramic capacitors.

The RMS current of the input capacitor is approximately:

$$I_{CIN_RMS} = I_{OUT}\sqrt{D(1-D)}$$

From the equation, it can be seen that the highest RMS current occurs when D is 0.5:

$$I_{CIN_RMS} = \frac{1}{2} I_{OUT}$$

Choose the capacitor with RMS current rating higher than 1/2 I_{OUT}

The power dissipation on the input capacitor can be estimated with the RMS current and the ESR resistor. Electrolytic or tantalum capacitors can also be used, but due to their significantly higher ESR, a small size ceramic capacitor should be placed as close to the IC as possible.

The voltage ripple on the input capacitor, neglecting the ESR impact, can be calculated as:

$$\Delta V_{CIN} = \frac{I_{LOAD}}{f_{SW} \cdot C_{IN}} \cdot \frac{V_{OUT}}{V_{IN}} \cdot (1 - \frac{V_{OUT}}{V_{IN}})$$

Output Capacitor Selection

An output capacitor is required to obtain a stable output voltage. To minimize the output voltage ripple, ceramic capacitors should be used, and the ripple voltage can be estimated as:

$$\Delta V_{OUT} = \frac{1}{8} \cdot (1 - \frac{V_{OUT}}{V_{IN}}) \cdot \frac{V_{OUT}}{L} \cdot \frac{1}{(f_{SW})^2 \cdot C_{OUT}}$$

If electrolytic or tantalum capacitors are used, the ESR will dominate the output voltage ripple:

$$\Delta V_{OUT} = (1 - \frac{V_{OUT}}{V_{IN}}) \cdot \frac{V_{OUT}}{f_{SW} \cdot L} \cdot R_{ESR}$$

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PCB Layout Recommendation

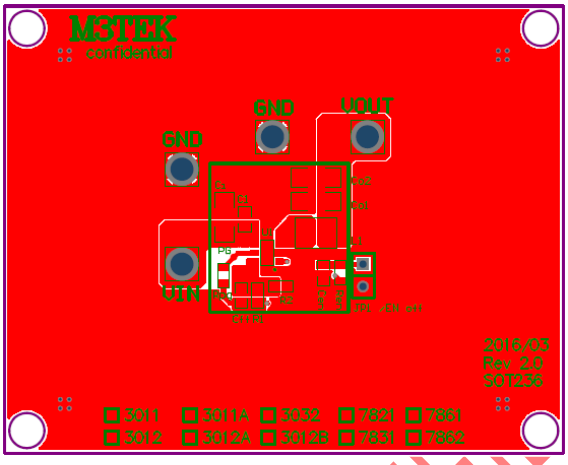
The physical design of the PCB is the final stage in the design of power converter. If designed improperly, the PCB could radiate excessive EMI and contribute instability to the power converter. Therefore, following the PCB layout guidelines below can ensure better performance of MT8133.

(1). The loop (Vin-SW-L-Cout-GND) indicates a high current path. The traces within the loop should be kept as wide and short as possible to reduce parasitic inductance and high-frequency loop area. It is also good for efficiency improvement.

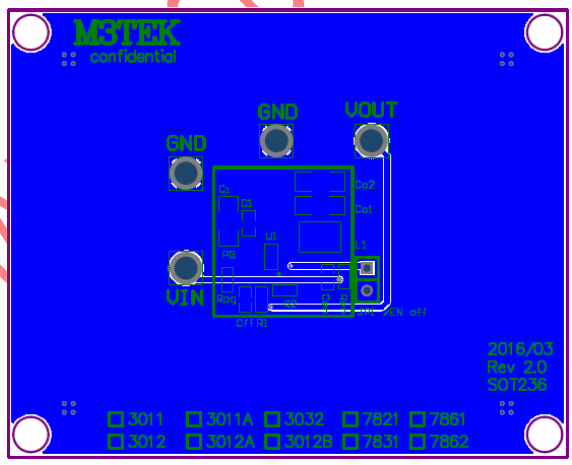
(2). Input capacitor as close as possible to the IC Pins (Vin and GND) and the input loop area should be as small as possible to reduce parasitic inductance, input voltage spike and noise emission.

(3). Feedback components (R_1, R_2 and C_{FF}) should be routed as far away from the inductor and the SW Pin as possible to minimize noise and EMI issue.

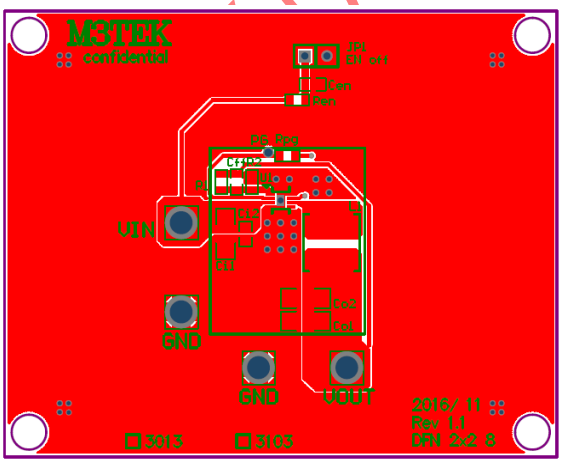
Top Layer (SOT235L/6L)



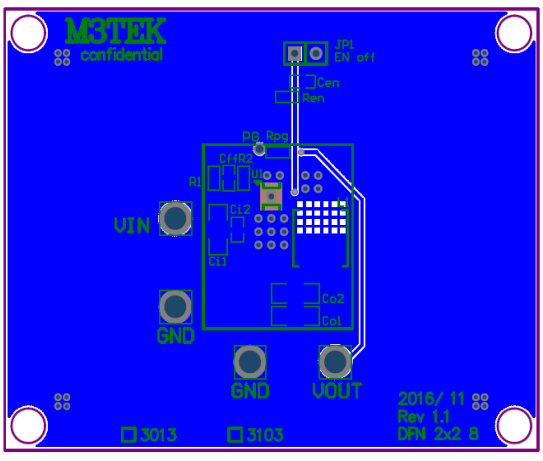
Bottom Layer (SOT235L/6L)



Top Layer (DFN2x2_8L)

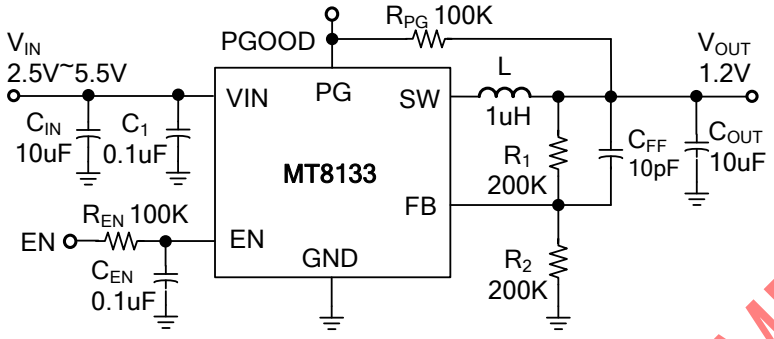


Bottom Layer (DFN2x2_8L)



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MT8133 Application Schematic



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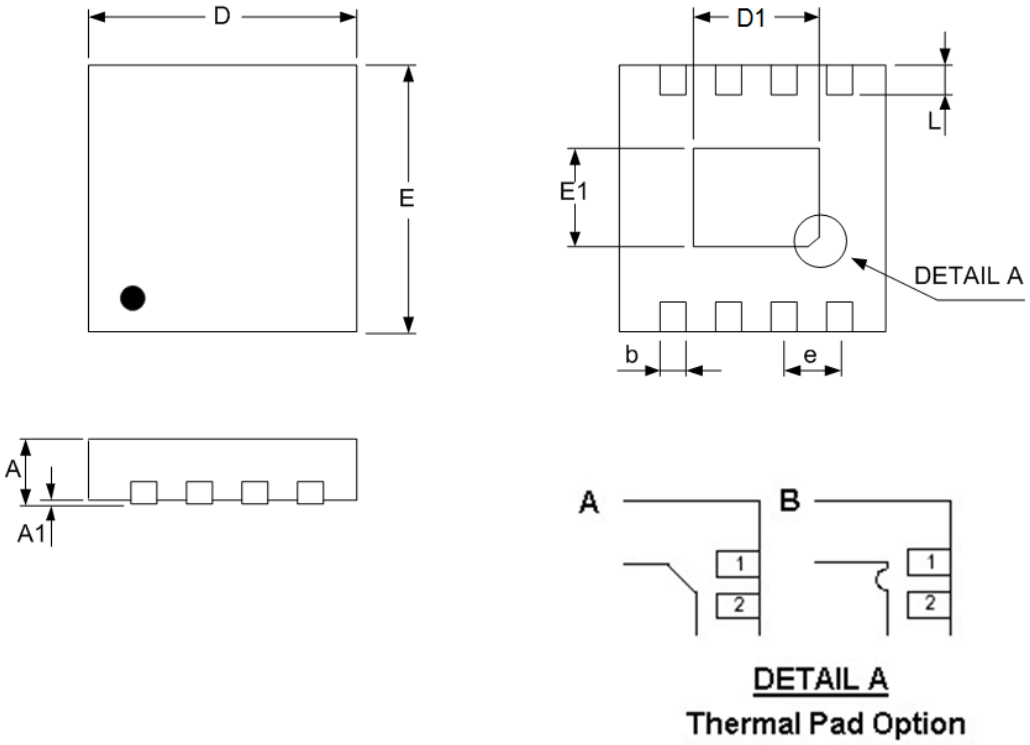
EVB BOM List

Qty	Ref	Value		Description	Package
1	C_{IN}	10μF		Ceramic Capacitor, 10V, X5R	0805
1	C_{OUT}	10μF		Ceramic Capacitor, 10V, X5R	0805
2	C1, C_{EN}	0.1μF		Ceramic Capacitor, 10V, X5R	0603
1	C_{FF}	10pF		Ceramic Capacitor, 10V, X5R	0603
1	L	1uH (0.47uH~1.5μH)		Inductor, GSTM4020P-1R0M, 22mΩ, 8.5A	SMD
1	R1	Vout=3.3V	200KΩ	Resistor, ±1%	0603
		Vout=2.5V	240KΩ		
		Vout=1.8V	200KΩ		
		Vout=1.2V	200KΩ		
		Vout=1.0V	100KΩ		
1	R2	Vout=3.3V	43KΩ	Resistor, ±1%	0603
		Vout=2.5V	75KΩ		
		Vout=1.8V	100KΩ		
		Vout=1.2V	200KΩ		
		Vout=1.0V	150KΩ		
2	R_{PG}, R_{EN}	100KΩ		Resistor, ±1%	0603
1	Power IC	MT8133		Step-Down DC/DC Converter	DFN2x2_8L SOT23_5L/6L

Package Information

DFN 8L 2x2mm Outline Dimensions

Unit: inches/mm



DETAIL A

Thermal Pad Option

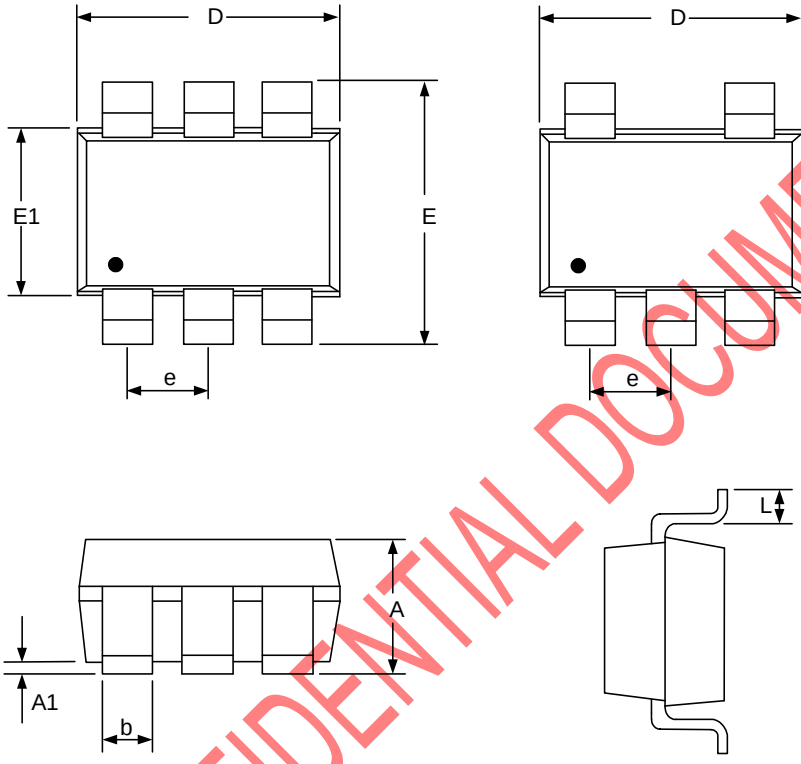
SYMBOLS	MILLIMETERS		INCHES	
	MIN.	MAX.	MIN.	MAX.
A	0.70	0.80	0.028	0.031
A1	0.00	0.05	0.000	0.002
b	0.18	0.30	0.007	0.012
D	1.90	2.10	0.075	0.083
D1	1.10	1.65	0.043	0.065
E	1.90	2.10	0.075	0.083
E1	0.50	1.10	0.020	0.043
e	0.50		0.020	
L	0.25	0.45	0.010	0.018

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Package Information

SOT23_5L and SOT23_6L Outline Dimensions

Unit: inches/mm

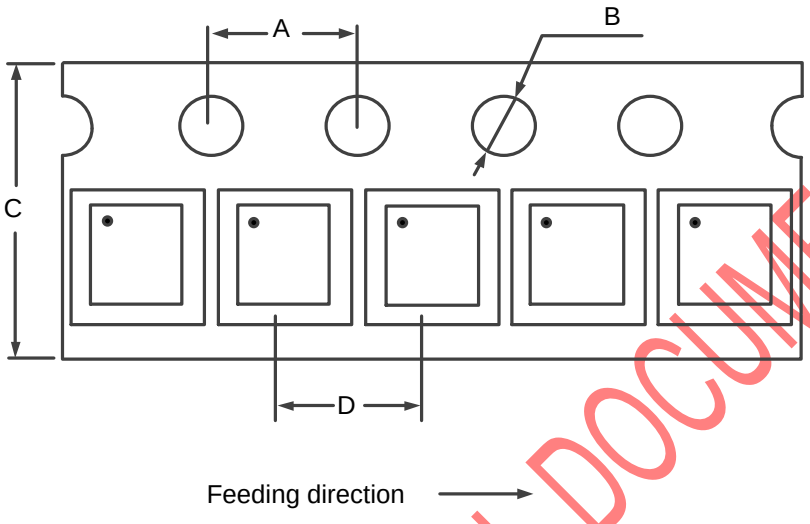


SYMBOLS	MILLIMETERS		INCHES	
	MIN.	MAX.	MIN.	MAX.
A	0.89	1.45	0.035	0.057
A1	0.00	0.15	0.000	0.006
b	0.30	0.50	0.012	0.020
D	2.70	3.10	0.106	0.122
E1	1.40	1.80	0.055	0.071
e	0.95 BSC		0.037 BSC	
E	2.60	3.00	0.102	0.118
L	0.30	0.60	0.012	0.024

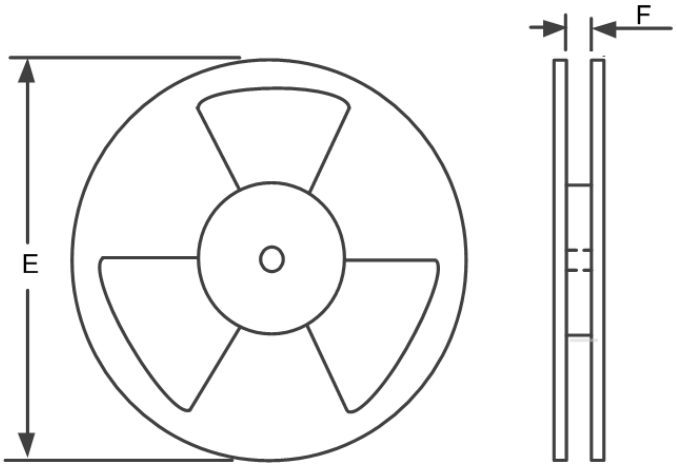
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Carrier Tape & Reel Dimensions (Packing Spec DFN2x2_8L)

1. Orientation / Carrier Tape Information:



2. Reel Information:



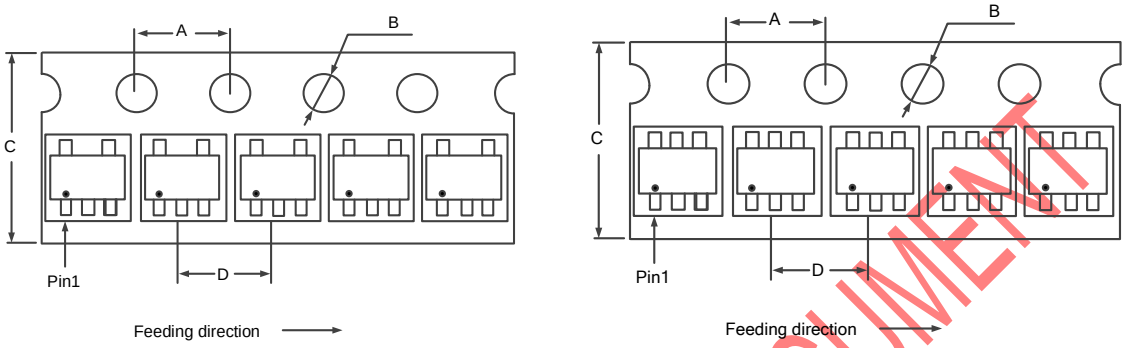
3. Dimension Details:

PKG Type	A	B	C	D	E	F	Q'ty/Reel
DFN2x2	4.0 mm	1.5 mm	8.0 mm	4.0 mm	7 inches	9.0 mm	3,000

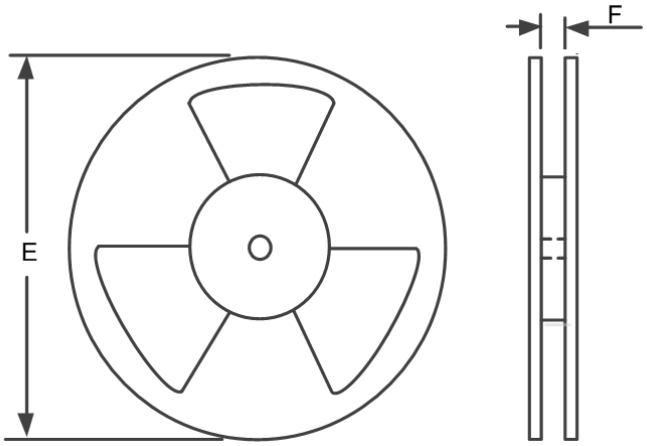
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Tape & Reel Carrier Dimensions

1. Orientation / Carrier Tape Information: SOT23_5L and SOT23_6L



2. Reel Information:



3. Dimension Details:

PKG Type	A	B	C	D	E	F	Q'ty/Reel
SOT23_5L/6L	4.0 mm	1.5 mm	8.0 mm	4.0 mm	7 inches	9.0 mm	3,000

M3TEK CONTROLLED DOCUMENT

Reflow Profile

Classification Of IR Reflow Profile

Reflow Profile	Green Assembly
Average Ramp-Up Rate (Ts_{min} to Tp)	1~2℃/second, 3℃/second max.
Preheat & Soak -Temperature Min(Ts_{min}) -Temperature Max(Ts_{max}) -Time(ts_{min} to ts ts_{max})	 150℃ 200℃ 60~120 seconds
Time maintained above: -Temperature(T_L) -Time(t_L)	 217℃ 60~150 seconds
Peak Temperature(Tp)	See Classification Temp in table 1
Time within 5℃ of actual Peak Temperature(tp)	30 seconds max.
Ramp-Down Rate	6℃/second max.
Time 25℃ to Peak Temperature	8 minutes max.

* Tolerance for peak profile Temperature (T_p) is defined as a supplier minimum and a user maximum.
** Tolerance for time at peak profile temperature (t_p) is defined as a supplier minimum and a user maximum.

Table 1. Pb-free Process – Classification Temperatures (Tc)

Package Thickness	Volume mm^3 <350	Volume mm^3 350-2000	Volume mm^3 >2000
<1.6 mm	260 ℃	260 ℃	260 ℃
1.6 mm – 2.5 mm	260 ℃	250 ℃	245 ℃
2.5 mm	250 ℃	245 ℃	245 ℃

Note: For all temperature information, please refer to topside of the package, measured on the package body surface.

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